Status	Finished
Started	Sunday, 25 October 2026, 2:45 PM
Completed	Sunday, 25 October 2026, 2:50 PM
Duration	5 mins 18 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Merging
- c. Searching
- d. Sorting

The correct answer is:
Complexity

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Merging two datasets
- c. Arranging data in a specific order
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are compared element by element
- c. They are discarded
- d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **6**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys

b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

c. A sorting algorithm is stable if it preserves the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort

b. Merge Sort

c. Insertion Sort

d. Quick Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass

b. It avoids the need for recursion

c. It only works on small datasets

d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It complicates the selection of items
c. It makes it easier to select items based on their relationship to the rest of the items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **11**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort is difficult to implement
c. Bubble Sort is less efficient for large lists
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Heap sort
c. Merge sort
d. Quick sort

The correct answer is:
Merge sort

Question **13**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It complicates the analysis process
c. It makes the analysis slower
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **14**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It bubbles up the largest element in each pass
c. It divides the list into sublists
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **15**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It does not require recursion
c. It is less efficient than bubble sort
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach